

Finlay Cooper

finlay.cooper@rogers.com • (647) 963 2667 • LinkedIn • GitHub • Portfolio

Education

Western University, London ON

June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**

- **Core Coursework:** Algorithms & Data Structures; Operating Systems; DBMS; Networking; Web Technologies; Information Security; Machine Learning; Cloud Computing; Software Testing & Maintenance.

Experience

Software Developer Intern, BroadGrain Commodities — Toronto, ON

Summer 2025

- **Owned full product lifecycle** as sole developer, defining requirements with plant managers, managing the backlog, and shipping iterative releases under real operational deadlines.
- Engineered a **schema-driven backend factory** to auto-generate CRUD controllers, routes, and validation from a single schema, reducing 100+ files to 3 and enabling zero-code endpoint creation.
- Integrated **Microsoft Graph API + Azure AD** to automate SharePoint document delivery with token caching, retries, and exponential backoff, eliminating manual report distribution entirely.
- Built **10 PDF reporting services** (JSReport, Handlebars, digital signatures, multi-photo support), replacing paper-based compliance workflows across 8 categories.
- Integrated **LLM tooling (MCP)** for automated legacy data migration and schema derivation; deployed to on-prem infrastructure (**pm2/IIS**) and authored runbooks for successor handoff.
- **Reduced processing time by 15% (refactoring) and 75% (digitization)** across reporting workflows.

Projects

GTA Management System — Capstone | control-alt-elites.netlify.app

2025–2026

- Built a **full-stack platform** replacing spreadsheet-based GTA hiring at Western Engineering, **deployed department-wide** across 100+ TAs, professors, and committee members.
- Designed a **configurable matching algorithm** combining skill match, preferences, course history, and review data with adjustable weights, executing assignments across 4 priority waves.
- Developed **78+ REST API endpoints** with role-based access control (student/professor/committee), Zod validation, and ownership enforcement; fully tested via automated bash suite.
- Implemented a **dry-run assignment system** enabling preview, weight tuning, and manual overrides before any assignments are committed.

Soi Ram — Freelance Restaurant Website | soiram.netlify.app

2025–2026

- Delivered a **production-grade Next.js / TypeScript / Tailwind site** with dynamic theming, fluid `clamp()` typography, and data-driven menu rendering; managed the full client lifecycle from design to deployment.
- Integrated **Toast POS ordering** alongside UberEats and DoorDash to enable commission-free direct orders with a unified receipt-style UI.
- Designed a **retro 80s brand identity** with custom typography, mobile-first layout, and SEO optimization.

Fracture Protocol — 2-Person Unity Roguelike

2025–2026

- Engineered a **roguelike arena shooter** in Unity/C# with 77 scripts, 3 boss encounters, 12 elite enemy types, and a **32-modifier progression system** with 5 rarity tiers and diminishing returns on stacking.
- Implemented **coroutine-driven boss AI** for 3 distinct encounters, running concurrent behaviors (fire patterns, drone phases, respawn timers) with phase transitions at 66%/33% HP and 0.8s visual telegraphing.
- Architected a **modifier dispatch system** using string IDs queried at runtime, enabling dynamic effect composition across 32 modifiers without hardcoded branches.
- Managed **Git version control** in a 2-person concurrent Unity workflow, coordinating development across scripts, scenes, and prefabs.

Skills

Languages & Frameworks: TypeScript/JavaScript, Java, C#, Python, C++, SQL; React/React Native, Next.js, Node.js, Express.js, Swift/SwiftUI, Unity

Tools & Platforms: MySQL, Firestore, PostgreSQL, MongoDB; Prisma ORM; Azure/SharePoint, IIS, Nginx, Docker, Git, CI/CD; Networking (IP, Routing & Switching)

Spoken Languages: English, French